Project 2: Active Learning for Text Classification

Human-Centric Artificial Intelligence

In this project, you are asked to implement some active learning methods for text classification. The project is divided into three parts.

1 Preliminaries: Dataset

The task for the project is to train a classifier for sentiment analysis on the IMBD 50k movie reviews dataset. The dataset contains a total of 50k movie reviews extracted from IMBD. The goal is to predict whether a review is positive or negative.

The dataset is split into a training set and a test set. The test set will be used only for validation of the proposed methods. The training set is labeled, but for the active learning part, you can ignore these labels.

2 Description of the tasks

2.1 Task 1: Supervised learning model for text classification

Your first task consists in training a classification model for the texts. This model will be first trained on the full training dataset, including labels, in a passive manner.

Task 1

Propose a classifier for text classification. This classifier must be made up of two distinct components: a module for the representation of the text (encoding a text into a real vector), and a classification module. Train your classifier on the whole labeled training dataset, and report the test accuracy.

This step is very important, since this classifier will provide the target score for the active learning part: your hope is to get a similar score from a very small dataset as with the complete dataset.

Remarks:

- For the representation, you can use any representation you want, including (but not restricted to) TF-IDF, bag-of-words, word2vec, Latent Dirichlet Allocation, or train your own representation. If you decide to train your own representation, either use the IMDB dataset without labels, or use another labeled dataset of your choice.

- In the project page, have a first part where you describe your classifier. Have a button from which to launch the training of this classifier, but also load a pre-trained representation module and classifier (for instance using `pickle`).

2.2 Task 2: Pool-based Active learning

From now on, we consider that the labels are not available during the training step.

Task 2

Implement various active learning strategies in a pool-based setting. You can test several utility functions, and also mixtures. Make the choice of the utility functions interactive for a better testing.

Task 3

Test the implemented active learning either on simulated users (using the available labels as the user's response) or by proposing a simple labeling interface. Think of a termination condition for the active learning training, and display the score.

Remark: Use the same representation as in Task 1, and the same model of classifier. Do not used the pre-trained classifier.

2.3 Possible extensions

If you want to go further, you can implement any of these two extensions:

- **Batch AL:** Implement a method to query not one, but several points at the same time.
- **Stream-based AL:** Read the data from the dataset one by one, and implement a method to decide online whether to label or not this point. You can implement a sliding window strategy, but do not store the whole dataset!